



Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

Email: support@efyindia.com
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MASTER BOOT RECORD	INTER-PARTITION GAP	VOLUME BOOT RECORD	PARTITION No. 1	VOLUME BOOT RECORD	PARTITION No. 2
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Fig. 1: Partitioning of the hard drive

DOS BOOT SECTOR	FAT No. 1	FAT No. 2	ROOT DIRECTORY	USER DATA AREA
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Fig. 2: FAT 12/16 structure

DOS BOOT RECORD	RESERVED SECTORS	COPY OF DOS BOOT RECORD	RESERVED SECTORS	FAT No. 1	FAT No. 2	USER DATA
32 SECTORS						

Fig. 3: FAT 32 structure

or directory has at least one entry in MFT, which is called file record and its default size is 1024 bytes. The first 42 bytes are fixed for MFT entry header and the rest of the entry stores attributes, which have a small data structure. Data is stored in the cluster, which is a small disk space. Every cluster in NTFS has a logical cluster number (LCN). The cluster number starts with zero.

There are also metadata files like \$boot, \$logfile, \$badclus, and \$volume that describe the file system. These metadata files can be explored with the help of Winhex software. MBR signature of NTFS is 0x07. In NTFS, the maximum file size is 2^{64} bytes, maximum number of files is $2^{32}-1$ and maximum length of a file name is 255 characters.

One important feature of NTFS is alternate data streams (ADS), which can be added with any file by the command `c:\notepad anyfile.txt:hide`. Here, any hidden data can be typed in with the text file anyfile.txt:hide. Hidden data may be of 100 kB but the size of anyfile.txt does not get changed. Many hackers use this ADS feature to hide information while uploading files onto the victim's computer. Many times, when a computer is saving a file on the hard disk, it cannot allocate the exact number of clusters to save the file. The last cluster will remain half filled and that's where a hacker can hide information.

System internal commands can be used to unearth this data.

Software for forensics

Investigators need software tools for file viewing, disk imaging, uncompressing files, identifying known files, performing string searches and accessing file metadata. WinHex software (www.x-ways.net) is used for making hard disk images, viewing files and recovering deleted files. ProDiscover (www.techpathways.com) is a forensic software, which is used for disk imaging and file signature analysis to recover deleted files. This software is suitable for analysis of FAT, NTFS and Sun Solaris UFS. It also displays ADS streams in the file system. Encase is another popular tool used in the forensic field. This software safely previews a disk before acquisition and provides a virtual boot disk image using VMWare to allow a first-hand view of the system.

You can use SafeBack, Ghost, Mareware and FTK software for disk imaging. Process Explorer, AutoRuns, WireShark, MD5Sum and Event-CombMT are other free software for digital forensic purposes. These software tools will help you in protecting your system, and empower you to watch out for hackers' data on your computer. ●

The author works as an information security consultant at MIEL eSecurity Pvt Ltd, Mumbai